

Reproducibility Checklist

1. General Paper Structure

- 1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) [yes — the differentiable execution \(ROM as weights, RAM as a soft tape, control flow as gates\) is described conceptually and formally](#)
- 1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) [yes — proven and measured claims are kept distinct from the explicitly qualitative XAI-feasibility claim](#)
- 1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) [yes — background on the VCS, the 6502, and ALE is cited](#)

2. Theoretical Contributions

- 2.1. Does this paper make theoretical contributions? (yes/no) [yes — we prove the soft \(differentiable\) forward execution equals the hard execution bit-for-bit at any finite temperature](#)

If yes, please address the following points:

- 2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) [yes — the finite-temperature and straight-through assumptions are stated formally](#)
- 2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) [yes — the soft-equals-hard equality is given as formal theorems](#)
- 2.4. Proofs of all novel claims are included (yes/partial/no) [yes — full proofs are in the supplementary material](#)
- 2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) [yes — proof sketches and intuitions accompany the formal results](#)
- 2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) [yes — citations to the underlying tools \(e.g., the straight-through estimator\) are given](#)
- 2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) [yes — the forward-equivalence theorem is verified quantitatively: all 64 games are byte-identical in RAM and pixel-identical on screen \(zero differing bytes and zero differing pixels, measured per byte and per pixel\), and the temperature-limit / fully-relaxed regime is measured on the full simulator; the surrogate gradients are additionally illustrated in the qualitative XAI study](#)
- 2.8. All experimental code used to eliminate or disprove

claims is included (yes/no/NA) [yes — the conformance harness and the soft-vs-hard verifier are included](#)

3. Dataset Usage

- 3.1. Does this paper rely on one or more datasets? (yes/no) [yes — the Atari 2600 cartridge ROMs / ALE games](#)

If yes, please address the following points:

- 3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) [yes — Atari/ALE is the platform on which deep RL was established; all 64 supported games are used](#)
- 3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) [NA — no novel datasets are introduced](#)
- 3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) [NA — no novel datasets are introduced](#)
- 3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) [yes — ALE and the Atari platform are cited](#)
- 3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) [yes — the standard ALE ROM set is publicly obtainable \(e.g., via ALE / AutoROM\)](#)
- 3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying (yes/partial/no/NA) [NA — all datasets used are publicly available](#)

4. Computational Experiments

- 4.1. Does this paper include computational experiments? (yes/no) [yes — bit-exact conformance sweeps, throughput benchmarks, and a gradient-based XAI study](#)

If yes, please address the following points:

- 4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) [NA — the paper introduces no learning algorithm with tunable hyperparameters; the emulator is a deterministic re-implementation, so there is no hyperparameter search or final-setting selection \(the fixed soft and benchmark settings are listed in the supplementary](#)

material)

- 4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) [yes — the code that prepares each run is included: the deterministic action-stream generator and the standard 60-NOOP+4-RESET boot; the ROMs are used unmodified](#)
- 4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) [yes — all experiment source is included \(attached for review; repository released on acceptance\)](#)
- 4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) [yes — released publicly on acceptance under the MIT License](#)
- 4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) [yes — every differentiable \(soft\) source in both ports \(jaxtari/jaxtari/diff/, jutari/src/diff/\) carries module- and function-level comments that name the paper construct each step implements—the five differentiable primitives \(one-hot read, softmax dispatch, sigmoid branch, straight-through estimator, distance-softmax relaxed read\), the forward-equivalence and temperature-limit theorems, the surrogate-gradient corollary, and the three execution modes—together with the XTARI routine each one mirrors](#)
- 4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) [yes — deterministic action streams and the Mnih-style random-NOOP-reset seeding are described and recorded](#)
- 4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) [yes — the supplementary *Throughput* section specifies the CPU \(Apple M1 Max\) and GPU \(NVIDIA GTX 1080 Ti, 11 GB, and Quadro RTX 5000, 16 GB, on an institutional Slurm cluster\) models and memory, plus the software stack and versions \(JAX 0.10.1 with CUDA 12, Python 3.12, Julia 1.12, Slurm 21.08.5\).](#)
- 4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) [yes — byte-identical RAM / pixel-identical screen \(conformance\) and env-steps per second \(throughput\) are defined and motivated](#)
- 4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) [yes — every reported result states its run count: GPU throughput is measured over ten timed runs per configuration \(and the CPU baseline over three\), and the conformance/exactness results are deterministic and reported from a single exact run \(zero variance by construction\)](#)
- 4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) [yes — GPU throughput is reported as mean \$\pm\$ standard deviation over ten timed runs per configuration in the supplementary material's GPU throughput table \(the run-to-run spread is below 0.2%\), and the exactness results are reported as exact distributions \(zero variance by construction\)](#)
- 4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) [no — no significance tests are used—exactness is deterministic and throughput is a benchmark, not a compared-method claim](#)
- 4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) [yes — the soft sharpness/temperature, batch sizes, boot sequence, and numeric scope are consolidated in a settings table in the supplementary material](#)